

Point of care compatible sample preparation buffer for direct SARS-CoV-2 detection in gargle samples

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INTRODUCTION

The SARS-CoV-2 pandemic increased public interest in fast sample analysis of saliva or gargle samples. The gold standard for saliva or gargle sample processing still includes expensive and time-consuming RNA extraction. Here we propose a fast and simple sample treatment buffer (RACE buffer) as alternative for gargling samples, which is PCR compatible and applicable for point of care detection methods such as loop-mediated isothermal amplification (LAMP). These LAMP reactions can be verified with the naked eye, suggesting usage of the RACE buffer for either laboratory routine or point of care testing.

KEY FACTS

- 10 min workflow
- 99% less expenses
- user friendly
- point-of-care compatible
- PCR and LAMP compatible

RACE BUFFER WORKFLOW FOR RT-qPCR

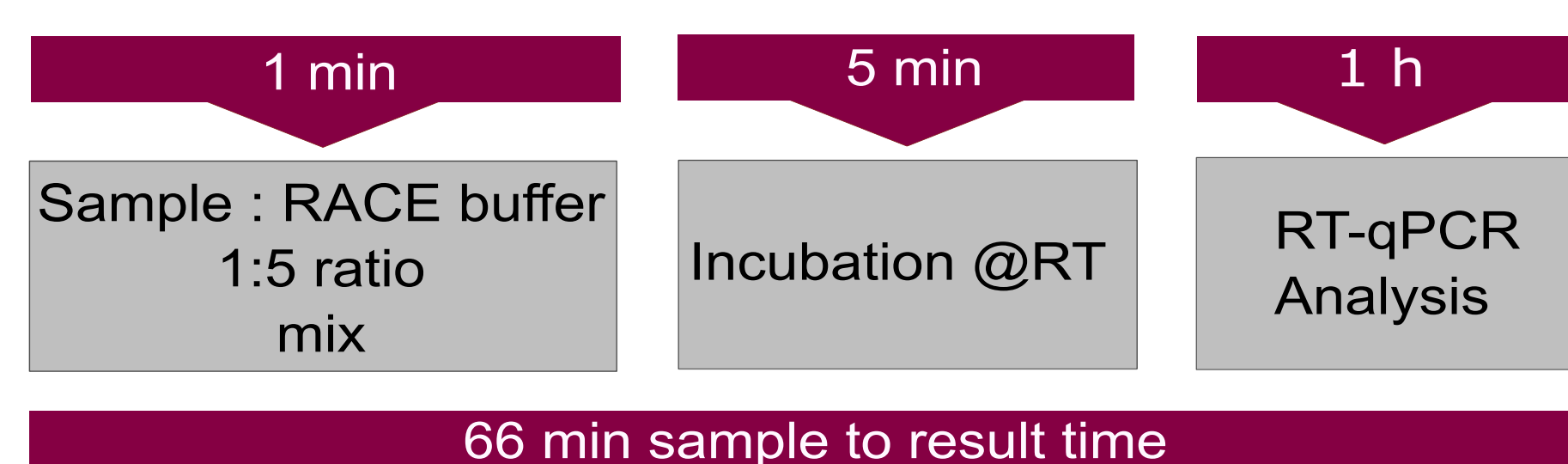


Figure 1: Rapid sample preparation workflow for RT-qPCR. Different sample treatment buffer components were tested for their PCR compatibility, resulting in the determination of the RACE buffer consisting of 2.5mM TCEP, 1% Triton-X100 reduced, 0.25mM EDTA and 5mM Tris-HCl. This buffer is mixed with gargle sample and incubated at room temperature for 5 min prior to direct PCR analysis. (RT: room temperature)

PERFORMANCE WITH CLINICAL GARGLE SAMPLES

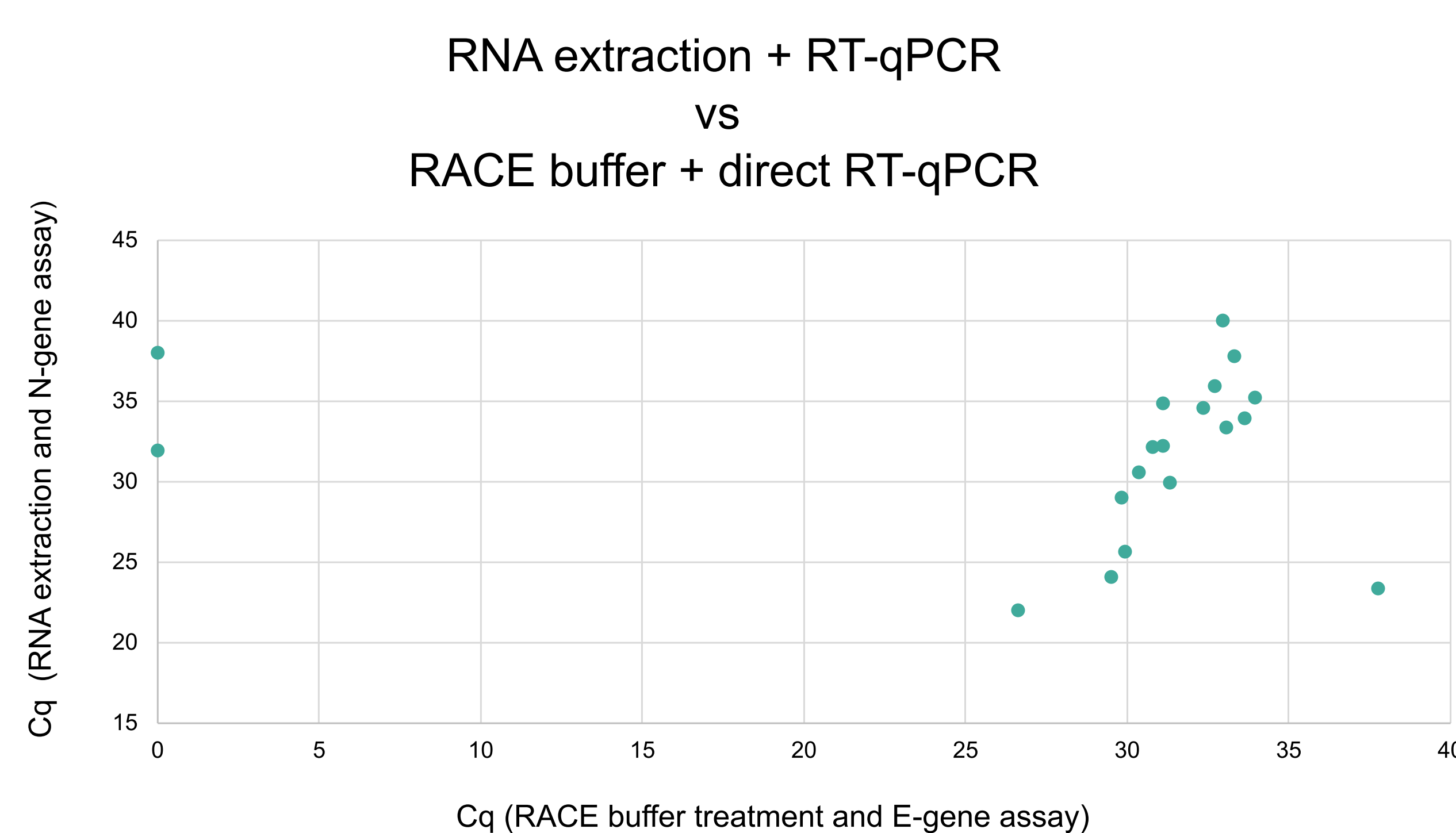


Figure 3: Comparison between the reference method (time consuming RNA extraction) and RACE buffer. Clinical samples were analyzed by RT-qPCR. The fresh samples were firstly analyzed with the reference method and afterwards temporarily stored at -80°C before testing the RACE buffer direct RT-qPCR workflow. Overall correlation between Cq values was observed. Two out of 19 samples showed a false negative result.

COMBATING RNA DEGRADATION IN GARGLE SAMPLES

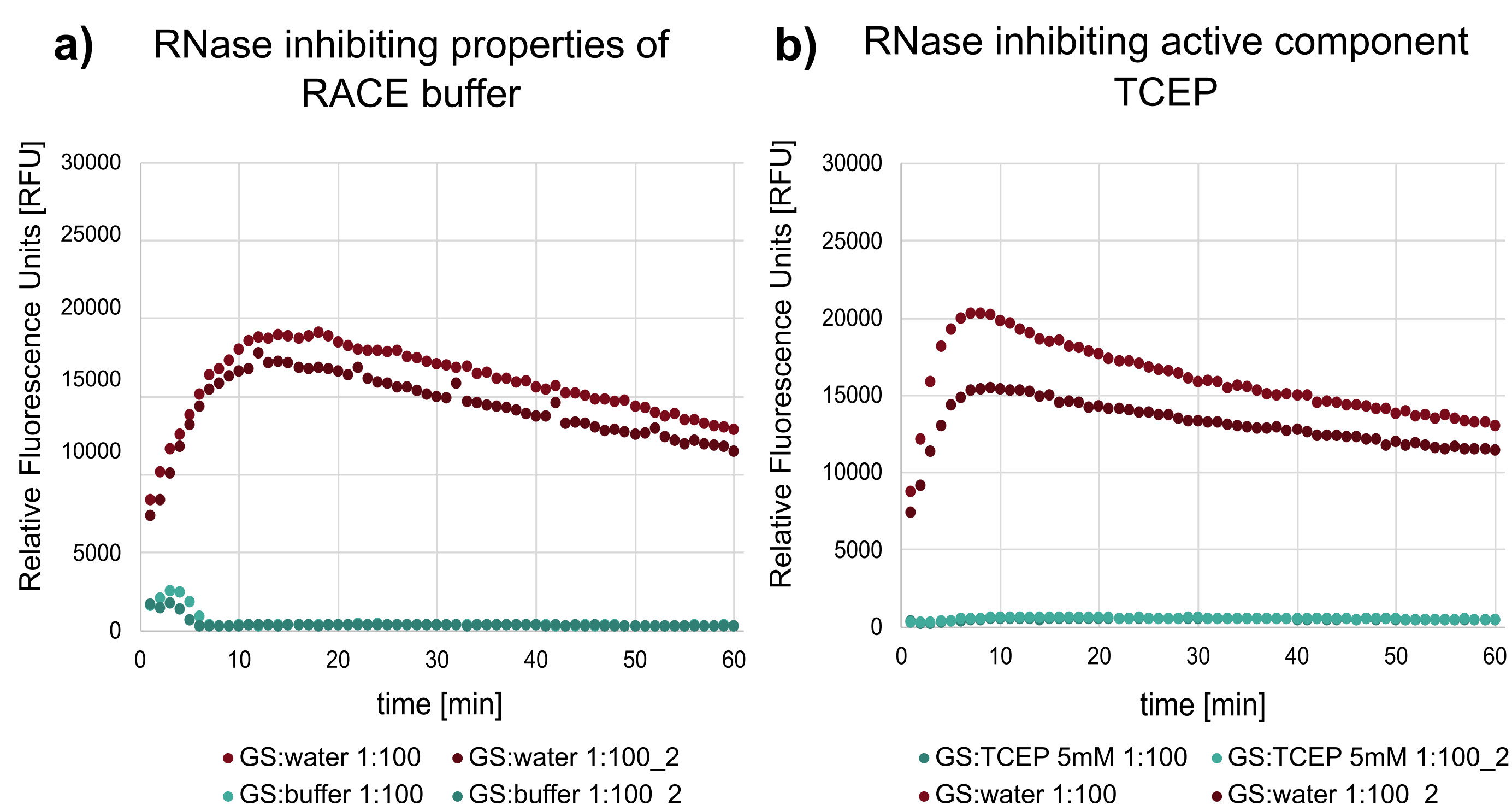


Figure 2: RNase inhibition. RNase activity in gargle samples (GS, red) vs inhibited activity in gargle samples treated with **a)** RACE buffer (turquoise) or **b)** 5mM TCEP (turquoise). The RACE buffer was shown to have RNase inhibiting effects, which will minimise RNA degradation in gargle samples. The RNase inhibiting component is TCEP (b). This leads to better sensitivity and increased sample stability.

COMPATIBILITY WITH RT-LAMP

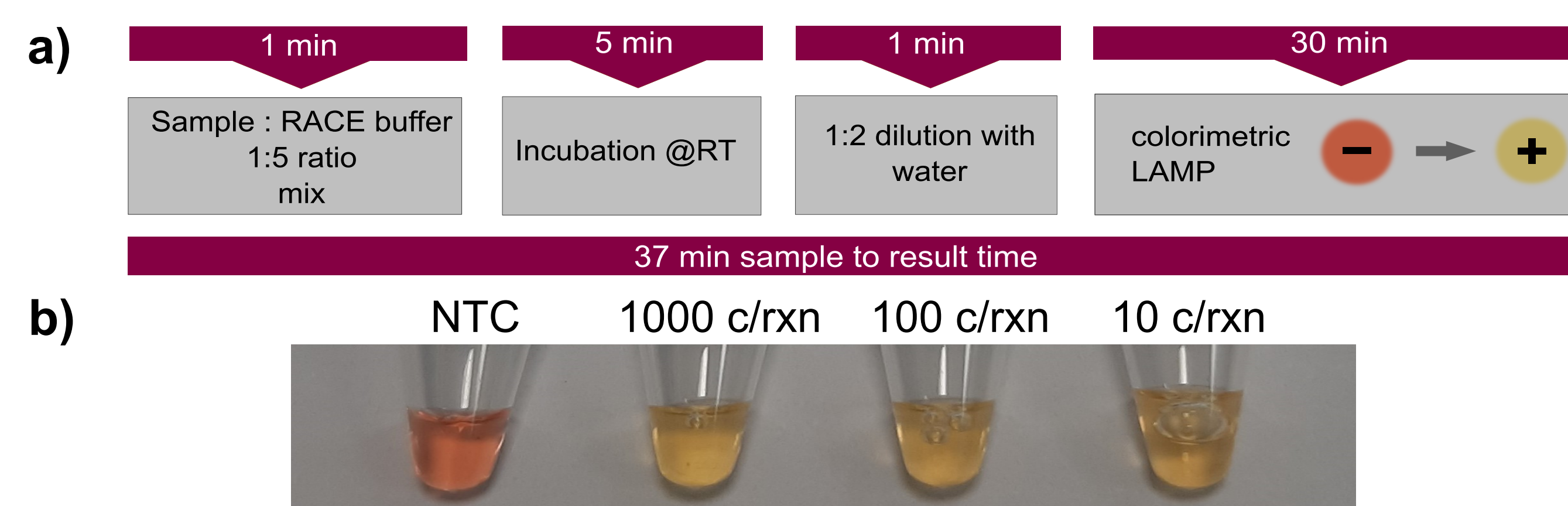


Figure 4: a) Rapid sample preparation workflow for RT-LAMP. b) RT-LAMP results for detection of SARS-CoV-2. RT-LAMP reactions were incubated at 65 °C for 30 min, if the amplification reaction is positive, colour change from orange to bright yellow can be seen with the naked eye. (RT: room temperature; c/rxn: copies per reaction)

component	final conc.
TCEP	2.5 mM
Triton X-100 reduced	1 %
EDTA	0.25 mM
TRIS-HCl	5 mM

CONCLUSION & OUTLOOK

With the developed RACE buffer the time required for sample preparation is reduced from 1.5h to about 10min and the material costs per sample are reduced from 3-5 EUR to 0.04 EUR. Moreover, in combination with RT-LAMP, reliable results with a limit of detection of 10 copies per reaction could be achieved within 37 minutes without any readout equipment. Furthermore, it's sample treatment performance got validated with direct RT-qPCR analysis of clinical gargle samples. In future perspective, this buffer could be used either in laboratory routine analysis of gargle samples as fast and cheap alternative to RNA extraction or in combination with point-of-care molecular diagnostic platforms.